

END-SEMESTER EXAMINATION (Summer), July 2026

Course Code: EAEPC303

Course Title: Probability Theory and Random Process

Duration: 03 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any two parts of the following		
1a	5 balls are to be selected randomly from a group of 5 blue and 10 red balls. (i) Find the probability that 2 blue and 3 red balls are selected. (ii) Find the probability that all selected balls are red.	5	CO3
1b	Two manufacturing plants produce similar parts. Plant-1 produces 1000 parts, 100 of which are defective. Plant-2 produces 2000 parts, 150 of which are defective. A part is selected at random and found to be defective. What is the probability that it has been produced by Plant-1?	5	CO3
1c	Let X be a random variable with distribution $f(x) = \begin{cases} K + \frac{x}{16} & 0 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$ (i) Find the value of K (ii) Find the probability $P(1 \leq X \leq 2)$	5	CO3
Q 2	Attempt any two parts of the following		
2a	Find the characteristic function of a normal random variable, $X = N(\mu; \sigma^2)$.	5	CO3
2b	The joint pdf of bivariate random variables (X, Y) is given by $f_{XY}(x, y) = \begin{cases} k(x + y) & 0 < x \leq 2, \quad 0 < y \leq 2 \\ 0 & \text{otherwise} \end{cases}$ where k is a constant (i) Find the value of k. (ii) Find the marginal pdfs of X and Y. (iii) Are X and Y independent? Explain.	5	CO3
2c	The joint pmf of bivariate random variables (X, Y) is given by $p_{XY}(x_i, y_j) = \begin{cases} \frac{1}{3} & (0, 1), (1, 0), (2, 1) \\ 0 & \text{otherwise} \end{cases}$ (i) Are X and Y independent? Explain. (ii) Are X and Y uncorrelated? Explain.	5	CO3
Q 3	Attempt any two parts of the following		
3a	Let X be a random variable with continuous pdf (probability density function) $f_X(x)$ and let $Y = X^2$. Find the pdf of Y.	5	CO3
3b	Determine the pdf of $Y = \sin(X)$, where X is a uniform random variable with pdf $f_X(x) = \begin{cases} \frac{1}{2\pi} & -\pi < x \leq \pi \\ 0 & \text{otherwise} \end{cases}$	5	CO3
3c	State and prove Chebyshev inequality.	5	CO1,2,3

Q 4	Attempt any two parts of the following		
4a	The autocorrelation function for a stationary ergodic process with no periodic components is given by, $R_{XX}(\tau) = 25 + \frac{4}{1 + 6\tau^2}$ Find mean value and variance of the process.	5	CO3
4b	Given the random process, $X(t) = A \cos(\omega t + \theta)$ If A and ω are constants and θ is a uniformly distributed random variable on the interval $(0, 2\pi)$. Is the process wide-sense stationary? Justify your answer.	5	CO3,5
4c	What is the autocorrelation function of random process? Explain the properties of autocorrelation function.	5	CO1,2
Q 5	Attempt any two parts of the following		
5a	State and prove three main properties of power-density spectrum.	5	CO1,2,5
5b	Derive the relationship between the autocorrelation function and the power density spectrum.	5	CO1,2,5
5c	Explain "white noise" along with its power density spectrum and autocorrelation function. Is "white noise" realizable? Explain your answer.	5	CO1,2